MAT1120

Def. Lie group is smooth mfd w/ gp structure w/ smooth operations.

ex. (Classical gps) $GL_n\mathbb{R}$ or $\mathbb{C}$

$SL_n\mathbb{R}$

$O(n)$

$SO(n)$

$U(n)$

$SU(n)$

$SU(2) = \{(z,w) : 1 = |z|^2 + |w|^2\}$ $\in \mathbb{C}^2$ " $S^3 \subset \mathbb{H}$ quaternion multiplication.

$$\begin{pmatrix} z & \bar{w} \\ \bar{z} & w \end{pmatrix} \longleftrightarrow z + wj, \quad z, w \in \mathbb{C}, \quad |z|^2 + |w|^2 = 1.$$

matrix exponential $e^X := \sum_{j=0}^{\infty} \frac{X^j}{j!}$

cges wrt. Hilbert–Schmidt norm

$$|X| = \left(\sum_{j,k=1}^{n} |X_{jk}|^2 \right)^{1/2}.$$

- $|X+Y| \leq |X| + |Y|$
- $|XY| \leq |X||Y|$ (from Cauchy–Schwarz)

1) $e^0 = I$

2) $(e^X)^* = e^{X^*}$

3) $(e^X)^{-1} = e^{X^{-1}}$

4) $XY = YX \Rightarrow e^X e^Y = e^Y e^X$ (special case of BCH formula)

5) $C \in GL_n(\mathbb{C}) \Rightarrow e^{CXC^{-1}} = Ce^XC^{-1}$

6) $X \in M_n(\mathbb{C}) \Rightarrow \frac{d}{dt} e^{tX} = Xe^{tX} = e^{tX}X$

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Lie algebras

Lie group $\xrightarrow[\text{tangent space at } e.]{}$ Lie alg

every $x \in \mathfrak{g}$ $\rightsquigarrow$ left-invariant vector field $\hat{x}$ i.e.

with $L_g : G \to G$, $h \mapsto gh$ then $(L_g)_* \hat{x} = \hat{x}$.

X, Y, Z vector fields. $[X,Y] = XY - YX$.

satisfies: 1) $[X,X] = 0$

2) $[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0$.

ex. $Sp_n(\mathbb{R})$ or $Sp_n(\mathbb{C})$

real or complex matrix gps s.t. $\Omega X \Omega^T = X$ where $\Omega =$

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MATH120 Lec2.

ex. $U(1) = S^1 \subset \mathbb{C}$.

$\exp : \theta \mapsto e^{i\theta}$.

ex. $O(n) = \{A \in GL_n : A^T A = Id\}$.

$A = I + \alpha$, α represents tangent space

$(I+\alpha)(I+\alpha^T) = I$

$\alpha + \alpha^T = 0 \rightsquigarrow \alpha$ skew symmetric.

ex. $GL_n(\mathbb{R}) \xrightarrow{\text{tangent space}} M_n \mathbb{R}$.

ex. $SL_n(\mathbb{R}) \xrightarrow{\text{tangent space}}$ trace zero matrices.

follows from claim: $\det(\exp A) = \exp(\operatorname{tr} A)$.

ex. $U(n) = \{A \in M_{n\times n}\mathbb{C} : A^\dagger A = Id\}$

$A = I + \beta \rightsquigarrow \beta + \beta^\dagger = 0$ so β skew Hermitian.

ex. $SU(n) = \left\{ \begin{array}{l} A^\dagger A = Id \\ \det A = 1 \end{array} \right\}$. skew Hermitian w/ tr 0.

ex. Pauli matrices $\sigma_1 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $\sigma_2 = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$

$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$\{i\sigma_j\}$ basis for Lie alg of $SU(2)$.

Lie alg of $SU(2) \cong$ Lie alg of $SO(3)$.

$\exists\, 2:1$ surj $SU(2) \to SO(3)$. $A \in SU(2) \rightsquigarrow \sigma_A(x) = AxA^{-1}$

kernel is $\pm I$

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Recall $sl(2,\mathbb{C})$ generators $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $X = \begin{pmatrix} & 1 \\ & \end{pmatrix}$, $Y = \begin{pmatrix} & \\ 1 & \end{pmatrix}$.

Thm 4.34 in Hall: Let (π, V) be f-d. rep of $sl(2,\mathbb{C})$ (not necc. irrep). Then (1) all eigenvals of $\pi(H)$ are $\in \mathbb{Z}$.

(2) if v eigenvec for $\pi(H)$ w/ eigenval λ then $\lambda \geq 0$
$\pi(X)v = 0$.

Pf (1). v eigenvector of $\pi(H)$ eigenval λ.

$\boxed{\text{* from Thm 4.32}}$ $\exists N \geq 0$ st $\pi(X)^N v \neq 0$, $\pi(X)^{N+1} v = 0$.

So $\pi(X)^N v$ is eigenvec of $\pi(H)$ w/ eigenval $\lambda + 2N$.

So $m := \lambda + 2N$ nonneg int.* $\Rightarrow \lambda$ integer

Special: if $\pi(X)v = 0$ then $N = 0$ so λ integer.

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